Vinicius Jameli

E-mail Personal website Updated: April 10, 2026

Education

Period	Degree	Institution	Area	Supervisor
05/2024 - Present	Ph.D	University of Toronto	Operations Research	Timothy C. Y. Chan
03/2018 - 12/2023	B.Sc., Hons	University of Campinas	Applied Mathematics	Aurelio R. L. Oliveira

Professional experience

Period	Company	Area	Role	Activities
07/2021 - 11/2021	Unisoma	Advanced analytics	Intern	Modeled, and solved large-scale optimization problems related to supply chain.

Teaching experience

Period	Institution	Activities
03/2023 - 12/2023	University of Campinas	Offered training for mathematics olympiads to middle and high school students.

Publications

[1] M. Moyano, V. J. Cabrera, K. Meza-Vasquez, M. Beltran-Villarroel, S. Muñoz-Bustos, G. Tello-Valenzuela, N. Ojeda-Ortega, and G. Cabrera-Guerrero. "Heuristics for the direct aperture optimisation in intensity modulated radiation therapy: a systematic literature review". In: *Artificial Intelligence Review* 59.1 (2026), pp. 1–20.

Patents and copyright

[1] T. Stanescu, V. J. Cabrera, K. Vat, and T. C. Y. Chan. *MR Magnetic Field Simulation Space*. Intellectual Property Disclosure OI2025-01712. Filed. Toronto, Canada, Jan. 2025.

Technical reports

[2] V. J. Cabrera. *First steps towards improving teletherapy inverse planning of the Boldrini Infant Center using external solvers*. Technical report. Unpublished. 2023.

[1] V. J. Cabrera. *First steps towards improving teletherapy inverse planning of the HC and CAISM using external solvers*. Technical report. Unpublished. 2023.

Conference abstracts

[3] V. J. Cabrera and A. R. L. Oliveira. "Planning IMRT with MLC: a linear approach with elastic constraints". In: *XXIX Scientific Initiation Congress of UNICAMP*. Campinas, Brazil: Galoá, Nov. 2021.

[2] V. J. Cabrera and A. R. L. Oliveira. "Optimization of external beam radiation therapy". In: *EURO 2022 Conference Handbook and Abstracts*. Espoo, Finland, July 2022.

[1] V. J. Cabrera and A. R. L. Oliveira. "Primal-dual interior point algorithm for the fluence map optimization problem". In: *Proceeding Series of the Brazilian Society of Computational and Applied Mathematics*. Vol. 9. 2022.

Presentations

[2] V. J. Cabrera and A. R. L. Oliveira. *Optimization of external beam radiation therapy*. 3° Optimization Workshop. Botucatu, Brazil, Aug. 2022.

[1] V. J. Cabrera and A. R. L. Oliveira. *Optimization of external beam radiation therapy*. Applied Mathematics Week (SEMAP). Campinas, Brazil, Aug. 2022.

Undergraduate projects

Year	Theme	Summary
2023	Thesis	Entitled *Primal-dual interior point algorithm for the fluence map optimization problem*, the work develops an interior point algorithm for a classical linear formulation of the FMO problem.
2023	Case study	The radiotherapy planning framework of the hospitals *Boldrini*, *CAISM*, and *HC - UNICAMP* were carefully studied, aiming to establish the first steps towards the implementation of external inverse planning tools in their *TPS*.

Awards

Year	Award title	Awarded by	Terms of award
2025-2026	Peri Family Graduate Scholarship	Department of Mechanical & Industrial Engineering, University of Toronto	Selected based on academic merit and commitment to pursuing innovative solutions in healthcare engineering.
2023	*Beatriz Neves* Award	Brazilian Society of Applied and Computational Mathematics (SBMAC)	Second best undergraduate research project in applied mathematics (national level).
2022, 2023	Best research project.	Institute of Mathematics, Statistics and Scientific Computing, Brazil	Best undergraduate research project in applied mathematics in both years (institutional level).
2017	Gold medal	Association Internationale Mathématiques sans Fronteriès	Received at the International Olympics Mathématiques sans Frontières.
2014	Silver medal	Brazilian Astronomy Society	Received at the 17º Brazilian Astronomy and Astronautics Olympiad.

Funding

Year	Purpose	Granted by
2024	Base funding package for Ph.D students	Department of Mechanical & Industrial Engineering, University of Toronto
2023	Fully funded support for the CIMPA school *Mathematical Modelling and Numerical Simulation in Medicine* in Cuba	Centre International de Mathématiques Pures et Appliquées (CIMPA)
2022	Travel support to present at the EURO conference in Helsinki, Finland	Institute of Mathematics, Statistics and Scientific Computing, Brazil
2022	Travel support to present at the EncBioMat conference in Rio de Janeiro, Brazil.	Institute of Mathematics, Statistics and Scientific Computing, Brazil
2022	Travel support to present at the Optimization Workshop conference in Botucatu, Brazil.	National Council for Scientific and Technological Development, Brazil
2020-2023	Research grant	National Council for Scientific and Technological Development, Brazil